

Class 21: Four Important Subspaces

We can use any any $m \times n$ matrix, A , to define two important subspaces.

1. $\text{Col } A = \text{set of all linear combinations of the columns of } A = \text{Span}\{\vec{a}_1, \dots, \vec{a}_n\}$

$\text{Col } A$ is a subspace of $\mathbb{R}^m$

dimension of $\text{Col } A = \# \text{ of pivots.}$

2. $\text{Nul } A = \text{set of solutions to the system } A\vec{x} = \vec{0}$
Subspace of $\mathbb{R}^n$ Dimension = $\#$ of free variables.

Practice problems

$$A = \begin{bmatrix} 1 & 3 & -4 & -3 & 1 \\ 0 & 1 & -3 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Find a basis for $\text{Col } A$.

Does $\text{Col } A = \mathbf{R}^5$?

Does $\text{Col } A = \mathbf{R}^3$?

Is the vector $\begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$ in $\text{Col } A$?

What is the dimension of $\text{Nul } A$?

Is the vector $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ in $\text{Nul } A$?

Find a basis for $\text{Nul } A$.

Definition: A linear transformation, T , from a vector space V to a vector space W is a function that assigns to each vector $\mathbf{x}$ in V a unique vector $T(\mathbf{x})$ in W , such that

T preserves addition and scalar multiplication

This means for all vectors $\mathbf{u}$ and $\mathbf{v}$ in V and all scalars c we have

$$T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$$

And
$$T(c\vec{u}) = cT(\vec{u})$$

Any linear transformation T from V to W defines two important subspaces.

1. Range of $T = \{T(\mathbf{x}) : \mathbf{x} \in V\}$ is a subspace of W .

2. Kernel of $T = \{\vec{x} \text{ in } V : T(\vec{x}) = \vec{0}\}$ is a subspace of V .

Key idea: If $V = \mathbf{R}^n$ and $W = \mathbf{R}^m$ and $T(\mathbf{x}) = A\mathbf{x}$ where A is some $m \times n$ matrix, then the range of T is the column space of A and the kernel of T is the null space of A .

Practice Problems #33

The set of all 2×2 matrices with real entries is a vector space denoted by $M_{2 \times 2}$. We can define a function $T : M_{2 \times 2} \rightarrow M_{2 \times 2}$ by $T(A) = A + A^T$

What is $T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right)$?

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 2a & b+c \\ b+c & 2d \end{bmatrix}$$

Prove that T is a linear transformation.

$$\text{Let } \vec{u} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ and } \vec{v} = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

$$T(\vec{u} + \vec{v}) = T\left(\begin{bmatrix} a+e & b+f \\ c+g & d+h \end{bmatrix}\right) = \begin{bmatrix} 2(a+e) & (b+f)+(c+g) \\ (b+f)+(c+g) & 2(d+h) \end{bmatrix}$$

$$= \begin{bmatrix} 2a+2e & (b+c)+(f+g) \\ (b+c)+(f+g) & 2d+2h \end{bmatrix} = \begin{bmatrix} 2a & b+c \\ b+c & 2d \end{bmatrix} + \begin{bmatrix} 2e & f+g \\ f+g & 2h \end{bmatrix}$$

$$= T(\vec{u}) + T(\vec{v}) \quad \text{so } T \text{ preserves addition}$$

Let s be any scalar

$$T(s\vec{u}) = T\left(\begin{bmatrix} sa & sb \\ sc & sd \end{bmatrix}\right) = \begin{bmatrix} 2sa & sb+sc \\ sb+sc & 2sd \end{bmatrix}$$

$$= \begin{bmatrix} s(2a) & s(b+c) \\ s(b+c) & s(2d) \end{bmatrix} = s \begin{bmatrix} 2a & b+c \\ b+c & 2d \end{bmatrix} = s \cdot T(\vec{u})$$

so T preserves scalar multiplication

practice continued

Find one matrix in the kernel of T . $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

$$\begin{aligned} T(A) &= A + A^T \\ &= \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

Describe all matrices in the kernel of T .

$$A = \begin{bmatrix} 0 & b \\ -b & 0 \end{bmatrix}$$

Describe all matrices in the range of T .

Any matrix in the range of T must have the form $\begin{bmatrix} 2a & b+c \\ b+c & 2d \end{bmatrix}$. The entry in row 1 column 2 must be the same as the entry in row 2 column 1.

If B is any matrix in the range of this T we must have $B = B^T$

Find a matrix A such that $T(A) = \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix}$

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ we must have $2a = 1$ $2d = 6$ and $b+c = 2$. There are infinitely many correct answers.

$$A = \begin{bmatrix} \frac{1}{2} & 2 \\ 0 & 3 \end{bmatrix} \text{ works. also } \begin{bmatrix} \frac{1}{2} & 17 \\ -15 & 3 \end{bmatrix}$$